Risk Assessment & Mitigation Plan

Supply Chain Performance Analysis: Shipment Delays, Cost Efficiency, Inventory Availability, and the Impact of Shortages on Patient Cases

1. Risk Register

Each risk lists the evidence found in the source file or in the project plan, its impact, the mitigation, and the responsible team member. Counts are calculated from Data_Final_project_IR.xlsx.

ID	Category	Risk	Evidence	Impact	Mitigation	Responsible
R01	Data	No late shipments and a single shipment mode	Sheet in has 46 valid scheduled/delivered date pairs: 44 delivered on the scheduled date, 2 earlier and 0 later. Shipment mode is Air in 45 rows and blank in 1; managed by and fulfill via each hold one value.	Delay rankings, Late Rate comparisons and shipment-mode comparisons carry no information.	Report schedule status as a single chart; build the shipment analysis on lead time, customs clearance, shelf life, vendor value and freight cost; do not rank by delay or mode.	Hossam - Technical Analytics Lead
R02	Data	Small groups	Countries: Nigeria 17, Haiti 8, Cote d'Ivoire 6, Congo DRC 6, Guyana 5, Afghanistan 3, Ethiopia 1. Vendors: 6 of 11 have 3 or more shipments; 5 vendors have 1 shipment each.	Percentages and medians on very small groups are unstable.	Documented minimum group size (5 shipments for countries, 3 for vendors); smaller groups appear as Other with their count; every visual shows n.	Ola - Data Quality & Review Lead
R03	Data	Numeric fields stored as text	weight (kilograms): 38 numeric and 8 text values (for example Weight Captured Separately, See ASN-...). freight cost (usd): 38 numeric and 8 text values (for example Freight Included in Commodity Cost, See ASN-...).	Freight/Value and Freight per Kg are calculated on 38 records; totals are wrong if text is forced to zero.	Safe numeric conversion; non-convertible text becomes NULL with a status flag; ratios use sums over valid rows; the excluded count is logged.	Yahia - Data Preparation & Testing Lead
R04	Data	Date fields containing status text	po sent to vendor date: 45 valid dates and 1 value Date Not Captured. pq first sent to client date: 25 valid dates and 21 values Pre-PQ Process.	Lead Time sample is 45 records; invalid dates would produce false durations if converted.	Parse valid dates only; keep the status text in a status column; derived date stays NULL; show n for every Lead Time result.	Yahia - Data Preparation & Testing Lead
R05	Data	Missing values	Shipments: customs_clearance_days blank in 1 row; expiry_date blank in 2; batch_number blank in 3; receiving_inspection_result blank in 6; dosage blank in 8; shipment mode blank in 1. Hospital layer: unmet_demand_qty blank in 1,678 inventory_daily rows; requested_qty blank in 99; dispensed_qty blank in 348; outcome blank in 39 patient cases; received_date blank in 194 purchase orders.	Rates and denominators change if blanks become zero.	Blanks remain NULL and are flagged; denominators use valid rows only; a decision is logged for every field in the cleaning log.	Ola - Data Quality & Review Lead
R06	Data	Text variants in shipment fields	customs_clearance_days holds two text values (3 days, ~6). receiving_inspection_result has 8 raw variants (for example Accepted, accepted, ACCEPTED, Pass, OK, Accepted - OK, Accepted with remarks). The country value CÃ´te d'Ivoire is mis-encoded (1 of 7 countries).	Customs and inspection results are split or lost; one country is mislabeled.	Documented mapping built from existing values only; approximate values keep a flag; raw values are preserved.	Yahia - Data Preparation & Testing Lead
R07	Data	Shipments that expired before delivery	expiry_date is earlier than delivered to client date in 2 of the 44 shipments with a valid expiry date (as much as 212 days earlier).	The shelf-life chart shows negative values; they may be data errors or real findings.	Flag the records, list them in a dedicated table, verify against the source; do not delete them.	Hossam - Technical Analytics Lead
R08	Data	Duplicate keys in hospital tables	inventory_daily: 10,793 rows and 10,631 unique record_id (162 extra). purchase_orders: 2,187 rows and 2,163 unique po_id (24 extra). patient_cases: 5,665 rows and 5,598 unique case_id (67 extra).	Double counting of stockouts, orders and cases.	Uniqueness test on each key; the duplicate key and its cause are identified before any deletion; every removed row is logged.	Hossam - Technical Analytics Lead

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R09	Data	Inconsistent identifiers	inventory_daily hospital_id has 17 raw variants (for example H_A, h-a, trailing spaces, and H-E in 23 rows); item_id has 93 raw variants against a 30-item catalog (I001 to I030). patient_cases key_drug_item_id has 36 raw variants and is used to link cases to items.	Joins to hospital and item dimensions fail or drop rows; the case-to-item link (Q19) is wrong.	Standardize hospital_id (H_A and h-a become H-A; H-E is flagged); flag invalid item codes; run a referential-integrity test before any join.	Yahia - Data Preparation & Testing Lead
R10	Data	Mixed date formats	inventory_daily date has mixed formats. patient_cases admission_date has text dates and 13 Excel serial numbers. purchase_orders order_date has 8 numeric values.	Wrong periods and wrong durations.	Parse with explicit formats; unparseable values stay NULL and flagged.	Yahia - Data Preparation & Testing Lead
R11	Data	Category variants	purchase_orders po_status has 14 raw variants that reduce to 4 statuses after case and spacing normalization; supplier_name_raw has 25 distinct raw names for 9 supplier_id values; patient_cases severity mixes 1/2/3 with Low/Moderate/High (247 blank); inventory_daily stockout_flag has 6 encodings (0, 1, Y, N, TRUE, FALSE).	Split categories and wrong counts; an unreliable stockout flag.	Mapping built from existing values only; counts before and after mapping are documented; the Stockout Day Flag is derived from unmet_demand_qty > 0 and compared with the source flag.	Ola - Data Quality & Review Lead
R12	Model	Two data families without a shared key	Sheet in shares no key with the hospital tables; the 9 relationships in schema_relationships are all between hospital-layer tables.	False inference that a historical shipment caused a hospital or patient result.	Two separate models; no relationship and no fact-to-fact link; no causal wording; Model view checked in the data-model task.	Aya - Data Dictionary & Requirements Lead
R13	Data	Synthetic hospital layer	The data dictionary describes four fictional hospitals, and the hospital layer is described in the data file as synthetic.	Results could be mistaken for medical evidence.	Present the layer as an illustrative analytical case; descriptive comparisons only; show group sizes and missing counts.	Ola - Data Quality & Review Lead
R14	Plan	Dependency chain before the dashboard build	The five dashboard page tasks in Week 5 depend on the cleaning, derived-field and data-model tasks in Weeks 2 to 4 (11 tasks).	A delay in cleaning or modelling delays every page.	Weekly exit gates; relationships and denominators tested in Week 4; page tasks start only after the models are validated.	Yahia - Data Preparation & Testing Lead
R15	Plan	Submission dates on Fridays	16 October, 6 November and 4 December 2026 are Fridays; project weeks run Sunday to Thursday.	Last-day slippage on DEPI submissions.	Deliverables are completed by the Thursday week-end milestone; Friday is used only for upload to GitHub and the LMS.	Yahia - Data Preparation & Testing Lead